Priority Based Selection Hyperparameter Analysis Across Three Forecasts

Original moderate stagger and wide stagger datasets

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14 October 2026

Main finding

The strongest broadly applicable setting for this maintenance-only benchmark is C30, Wide dense ladder. It reaches the highest measured worst-day availability on all three datasets. Its largest smoothness penalty relative to a separately tuned dataset winner is 0.56 percent. Dataset-specific settings remain preferable when the final smoothness improvement matters.

Dataset	Initial minimum	Default minimum and std	Best minimum and std	Best setting
Original	15	18 / 1.422087	18 / 1.401929	C57
Moderate stagger	16	19 / 0.911708	19 / 0.883419	C30
Wide stagger	17	20 / 0.616685	20 / 0.586797	C30

The study crosses 58 neighbourhood and selection settings with the actual original, moderate stagger, and wide stagger workbooks, producing 174 matched sweep cases. It also records repeat timings, reference checks, and any finalist budget extensions. All 203 run records use the same scoring dates and preserve maintenance opening states.

The moderate and wide datasets change initial maintenance ages and the required check mix. They are distinct problem instances. Their higher input availability is not an optimisation gain on the original forecast. All reported gains compare a configuration with the default on the same dataset.

The previous half-step recommendation reaches 19 available aircraft on wide stagger, compared with 20 for the default. It therefore loses on the primary objective on that dataset and should not be adopted universally from the original-only result.

This report extends the original-only study and replaces its recommendation as evidence for selecting settings across these three forecasts. It evaluates maintenance feasibility; operational constraints are audited separately.

Datasets and maintenance opening states

Measure	Original	Moderate stagger	Wide stagger
Aircraft	24	24	24
Supplied checks	278	278	277
A checks	144	136	137
B checks	72	71	70
C checks	38	40	37
D checks	24	31	33
Initial worst day	15	16	17
Initial population std	1.954052	1.716837	1.411357
Initial mean	21.679525	21.477973	21.488245
Checks already in progress	0	1	2
Full maintenance aircraft days	10248	11158	11144

The moderate input has three cohorts of eight aircraft with maintenance timelines advanced by one, three, or five model years. The wide input has six cohorts of four, advanced by one through six model years. A model year is 365 days. Cohort assignment uses the saved generation seed 42 and was not chosen by the optimiser.

The aged fleets can start at A, B, or C within the repeating A B A C A B A D sequence. The existing opening-state adapter retains each aircraft's saved sequence index and A, B, C, and D due clocks. Every check that started before the benchmark start of 21 January 2026 is frozen. The original dataset retains its production rebuild assumption that its opening A cannot precede the first due date.

Changing only FILE_PATH would incorrectly reconstruct the aged fleets as if all aircraft were at the original opening A position. The study instead reads .scratch/initial-maintenance-stagger/schedule-data.json and replays the correct opening state. Check counts, types, durations, and these opening clocks remain fixed within each dataset.

Unchanged A B C D rules respectively use durations of 14 28 84 126 days, intervals of 365 730 1460 2920 days, and early allowances of 42 70 70 70 days. These are domain inputs, not search hyperparameters.

All datasets have 24 aircraft, but their D counts and downtime differ. Mean availability should therefore not be compared across datasets as a pure measure of optimiser quality.

Matched dates objective and validation

All three inputs are scored from 2026-01-21 through 2038-01-18, inclusive, for 4381 daily observations. The internal coordinate origin is 2025-11-16, allowing checks that began before the scoring dates to retain their full intervals. The model horizon is day 4546 and includes the full ends of supplied checks crossing the forecast boundary.

The objective is the lexicographic tuple (minimum availability, negative population standard deviation, mean availability). A higher worst day outranks every smoothness gain. Scores are calculated with the same NumPy functions used by the production method, on the matched visible window. The algorithm never accepts a worse endpoint.

The earlier report used 4481 observations through 28 April 2038. The matched study uses the published 4381-day forecast through 18 January 2038. Checks crossing either boundary remain in the model. Their starts can change the maintenance counted inside the visible window, so the visible mean can change even though full check durations are conserved. Standard deviations from the two horizons must not be compared directly.

Verification of the study driver

All settings preserve legal starts, no overlap, the maintenance sequence, and due resets. Each endpoint is checked by complete rule replay. Daily availability is independently recomputed by the delta-array method and by aircraft occupancy masks. Incremental availability must match both exactly. Past checks, opening states, check counts, and durations are preserved, and every accepted score strictly improves.

Priority groups are cached within an iteration, unchanged maintenance intervals are omitted from availability updates, and clearly dominated proposals are screened conservatively. Retained proposals still receive the original exact score. Cached and reference neighbour lists match for every initial configuration and every final neighbourhood. Reference original-input runs also match complete trajectories and final schedule hashes after these performance changes.

The live source was reorganised under vn_priority_based_selection.py during execution. The sweep processes had already loaded the preceding version. Serial repetitions therefore load byte-identical frozen algorithm and adapter snapshots whose hashes match all sweep records. This prevents a source-version change from being mixed into the timing or quality comparison.

The 58-case sweeps run as independent processes. Their elapsed times are retained as audit observations but are excluded from performance comparisons. Finalist timing runs execute serially after all sweeps finish.

Best configurations within each dataset

Original

Setting	Minimum	Std deviation	Mean	Moves	End
C57	18	1.401929	21.707601	302	Local
C02	18	1.404088	21.705090	300	Local
C56	18	1.404088	21.705090	300	Local
C25	18	1.405517	21.705318	311	Local
C43	18	1.406008	21.704177	338	Local

C57 uses Half steps plus 42 days. Relative to the default, worst-day availability changes from 18 to 18, and standard deviation changes from 1.422087 to 1.401929. At the same worst day, this is a 1.42 percent smoothness improvement.

Moderate stagger

Setting	Minimum	Std deviation	Mean	Moves	End
C30	19	0.883419	21.469071	367	Local
C49	19	0.885310	21.471582	411	Local
C03	19	0.887635	21.471810	414	Local
C57	19	0.892207	21.470441	314	Local
C58	19	0.894978	21.469299	366	Local

C30 uses Wide dense ladder. Relative to the default, worst-day availability changes from 19 to 19, and standard deviation changes from 0.911708 to 0.883419. At the same worst day, this is a 3.10 percent smoothness improvement.

Wide stagger

Setting	Minimum	Std deviation	Mean	Moves	End
C30	20	0.586797	21.475006	291	Local
C49	20	0.597531	21.473636	378	Local
C19	20	0.603282	21.474777	293	Local
C20	20	0.603592	21.473180	323	Local
C18	20	0.606233	21.473180	341	Local

C30 uses Wide dense ladder. Relative to the default, worst-day availability changes from 20 to 20, and standard deviation changes from 0.616685 to 0.586797. At the same worst day, this is a 4.85 percent smoothness improvement.

Local means natural exhaustion of the tested improving neighbourhoods. It does not mean globally optimal smoothness. Cap means the trajectory still has unresolved convergence.

Individual shift magnitudes and budget limits

Shift in days	Original minimum and std	Moderate minimum and std	Wide minimum and std
1	17 / 1.654990 cap	18 / 1.344982 cap	18 / 1.082390 cap
2	17 / 1.528896 cap	19 / 1.170972 cap	19 / 0.887747 cap
3	17 / 1.500076	19 / 1.105460	19 / 0.841440
5	18 / 1.462199	18 / 1.095632	20 / 0.793431
7	18 / 1.469950	19 / 1.099823	19 / 0.805418
10	17 / 1.465157	17 / 1.070767	19 / 0.792254
14	18 / 1.481357	18 / 1.045514	19 / 0.778973
21	17 / 1.539122	18 / 1.061332	19 / 0.797280
28	17 / 1.565118	18 / 1.112248	19 / 0.843196
35	17 / 1.588813	17 / 1.157659	19 / 0.913454
42	17 / 1.608533	18 / 1.214228	19 / 0.946695
56	16 / 1.707803	17 / 1.341924	19 / 0.978886
70	15 / 1.942977	16 / 1.705900	17 / 1.363817

This comparison holds selection fixed and uses one symmetric pair of offsets. The same shift does not necessarily rank equally on all inputs because the maintenance overlaps and feasible windows differ. A wide step can pass over worsening intermediate schedules, but its proposed endpoint still has to improve the full score.

A cap-limited small-step run is not evidence that its neighbourhood converges at the reported worst day. The revised study uses an 800-move cap rather than assuming that a 400 cap resolves each ordering or step-size comparison. Exact prefix results on the convergence page show the consequences of smaller budgets without counting prefixes as independent runs.

The variable-neighbourhood ladder remains a search setting, rather than a maintenance rule. The early allowances remain 42 days for A and 70 for B, C, and D. Temperature, cooling schedules, and a search random seed are outside this deterministic method.

Ladder structure ordering and local changes

Setting	Structure	Original min and std	Moderate min and std	Wide min and std
C01	Default	18 / 1.422087	19 / 0.911708	20 / 0.616685
C02	Half steps	18 / 1.404088	19 / 0.898502	19 / 0.633267
C03	Dense pairs	18 / 1.410612	19 / 0.887635	20 / 0.607737
C04	Wide first	18 / 1.408616	19 / 0.896507	20 / 0.638608
C31	Order small first default	18 / 1.444352	19 / 1.058317	19 / 0.749929
C32	Order small first weekly	17 / 1.654990	18 / 1.344982	18 / 1.082390
C33	Order interleaved	18 / 1.415108	19 / 0.936329	20 / 0.671271
C34	Order wide then day	17 / 1.422981	19 / 0.946293	19 / 0.685575
C50	Remove half offsets at level 1	18 / 1.413199	19 / 0.916204	20 / 0.621479
C51	Remove half offsets at level 2	18 / 1.408279	19 / 0.903569	19 / 0.637076
C52	Remove half offsets at level 3	18 / 1.407177	19 / 0.899518	19 / 0.634347
C53	Remove half offsets at level 4	18 / 1.413100	19 / 0.905091	19 / 0.636160
C54	Half steps reversed	18 / 1.450569	19 / 1.064117	19 / 0.781272
C55	Dense pairs reversed	17 / 1.654990	18 / 1.344982	18 / 1.082390
C56	Half steps plus two days	18 / 1.404088	19 / 0.898502	19 / 0.633267
C57	Half steps plus 42 days	18 / 1.401929	19 / 0.892207	20 / 0.608959
C58	Half steps plus 21 days	18 / 1.406948	19 / 0.894978	20 / 0.650567

Half steps adds intermediate offsets inside each level. Dense pairs adds separate intermediate levels. These differ because the search restarts at its first level after every accepted move and takes the first neighbourhood containing an improvement. A union of offsets, a layered ladder, and a reversed ladder can follow different trajectories even when they use similar magnitudes.

The ablations restore one half-step level to its original pair while retaining the other levels. Local refinements add two-day finishing, forty-two-day initial movement, or twenty-one-day offsets within the first level. Their effects are conditional on the remaining ladder and should not be added together as independent contributions.

Complete ordered shift tuples for every setting are provided in the final catalogue. Some structures occur with more than one selection policy.

Selection policy interactions across the forecasts

ID	Ladder	Selection	Original	Moderate	Wide
C01	Default	first level	18 / 1.422087	19 / 0.911708	20 / 0.616685
C40	Default	all levels	18 / 1.427404	19 / 0.923071	19 / 0.626372
C41	Default	first improving	18 / 1.428337	19 / 0.898510	20 / 0.636755
C02	Half steps	first level	18 / 1.404088	19 / 0.898502	19 / 0.633267
C42	Half steps	all levels	18 / 1.407822	19 / 0.903063	20 / 0.657894
C48	Half steps	first improving	18 / 1.413428	19 / 0.910190	20 / 0.612600
C03	Dense pairs	first level	18 / 1.410612	19 / 0.887635	20 / 0.607737
C43	Dense pairs	all levels	18 / 1.406008	19 / 0.897231	20 / 0.626964
C49	Dense pairs	first improving	18 / 1.411036	19 / 0.885310	20 / 0.597531

First level finds the first priority level containing any improving candidate and selects the best candidate in it. All levels selects the best improving candidate over every priority level in the active neighbourhood. First improving stops at the first improving candidate in deterministic order. Priority ties follow start date, aircraft name, check index, and offset order.

All policies use the same acceptance rule, so all-level selection does not permit a lower worst day. Differences in final smoothness arise from different accepted paths. This crossed comparison tests whether a policy recommendation made on the original data also holds on the aged fleets.

Within this interaction matrix, original is strongest with C02, first level, reaching 18 available aircraft and standard deviation 1.404088.

Within this interaction matrix, moderate stagger is strongest with C49, first improving, reaching 19 available aircraft and standard deviation 0.885310.

Within this interaction matrix, wide stagger is strongest with C49, first improving, reaching 20 available aircraft and standard deviation 0.597531.

Convergence and iteration budget recommendations

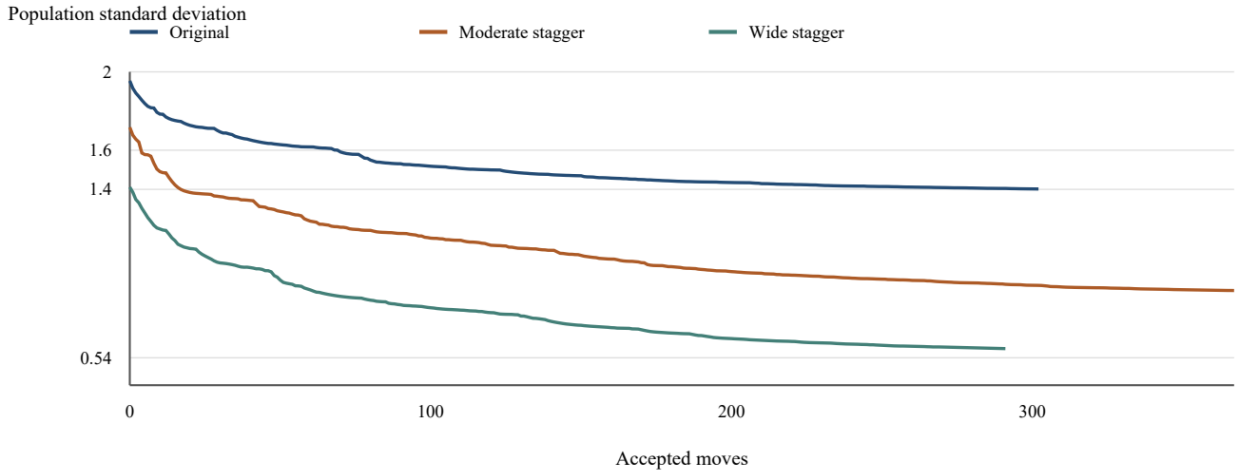


Figure 1 Best measured settings on each dataset. The horizontal axis is accepted moves, not elapsed time. Lower smoothness across datasets reflects different starting check mixes as well as optimisation.

Dataset	Budget	Default minimum and std	Best setting minimum and std
Original	20	17 / 1.754095	17 / 1.726734
Original	40	17 / 1.670376	17 / 1.652352
Original	80	17 / 1.575439	17 / 1.549478
Original	160	17 / 1.474137	18 / 1.457348
Original	200	18 / 1.446188	18 / 1.434867
Original	400	18 / 1.422087	18 / 1.401929
Original	800	18 / 1.422087	18 / 1.401929
Moderate	20	18 / 1.488380	17 / 1.384125
Moderate	40	18 / 1.340051	17 / 1.343284
Moderate	80	18 / 1.183961	18 / 1.190514
Moderate	160	19 / 1.071482	18 / 1.044012
Moderate	200	19 / 1.007132	19 / 0.979954
Moderate	400	19 / 0.911708	19 / 0.883419
Moderate	800	19 / 0.911708	19 / 0.883419
Wide	20	19 / 1.237274	18 / 1.097813
Wide	40	19 / 1.089956	19 / 1.000565
Wide	80	19 / 0.925512	19 / 0.834098
Wide	160	19 / 0.767225	20 / 0.693796
Wide	200	20 / 0.716456	20 / 0.637919
Wide	400	20 / 0.616685	20 / 0.586797
Wide	800	20 / 0.616685	20 / 0.586797

Checkpoint values are exact trajectory prefixes, not separate trials. A cap needs an extra iteration beyond the last accepted move to confirm natural exhaustion.

Shared settings and dataset specific recommendations

Setting	Original std penalty	Moderate std penalty	Wide std penalty	Worst penalty
C30	0.56%	0.00%	0.00%	0.56%
C49	0.65%	0.21%	1.83%	1.83%
C19	0.56%	2.06%	2.81%	2.81%
C18	0.82%	2.63%	3.31%	3.31%
C03	0.62%	0.48%	3.57%	3.57%
C57	0.00%	0.99%	3.78%	3.78%
C25	0.26%	1.60%	3.82%	3.82%
C44	0.51%	2.97%	4.08%	4.08%

C30 is the shared recommendation among naturally converged settings that retain each dataset's highest measured worst day. It minimises the largest relative smoothness penalty across the three inputs, using the average penalty to resolve a tie. It is a measured compromise on these forecasts, not a guarantee on a new fleet.

Dataset	Setting	Selection	Moves to stop	Suggested cap
Original	C57	first level	302	400
Moderate stagger	C30	first level	367	400
Wide stagger	C30	first level	291	400

Shared shift sets in order

Level 1 -56, +56 days

Level 2 -42, +42 days

Level 3 -28, +28 days

Level 4 -14, +14 days

Level 5 -7, +7 days

Level 6 -3, +3 days

Level 7 -1, +1 days

Shared selection first level

Shared max_iters 400. This covers the measured accepted-move counts with room to detect exhaustion.

The default remains the frozen baseline for method comparisons. A future annealing or basin-hopping comparison should face the tuned baseline on each input and use the same opening state, scoring dates, and operational assumptions.

Serial runtime measurements

Dataset	Setting	Median seconds	Range seconds	Accepted moves
Original	C01	14.09	14.04 to 14.85	259
Original	C02	33.33	31.70 to 33.92	300
Original	C57	32.16	30.37 to 32.29	302
Original	C30	29.22	29.12 to 29.94	332
Moderate	C01	21.62	21.27 to 23.24	347
Moderate	C30	14.69	12.83 to 32.88	367
Wide	C01	19.25	17.13 to 22.61	313
Wide	C30	28.55	27.93 to 30.60	291

Each timing group contains three serial executions after the parallel sweep processes have finished. Repeated endpoints and full schedule hashes must agree. These repeats estimate elapsed-time variability for a deterministic search; they do not provide a distribution of schedule quality.

Timing covers the search driver, copying, and trace recording. It excludes independent endpoint audits, neighbourhood equivalence checks after termination, operational audits, schedule exports, and report construction. The driver uses the same cached priority and exact-score screening implementation for every timing sample.

Host load and power management remain uncontrolled, so the medians and ranges describe this local machine. Sweep timings and timings from the earlier original-only report are excluded from the comparative table. The archived sweeps include reference implementations and concurrent execution that would otherwise confound a speed comparison.

Execution environment: Python 3.12.14, NumPy 2.3.5. No runtime claim is based on the fastest single run.

Operational audits and visible boundary effects

Dataset	Schedule	Max displacement days	Loose	Medium	Tight
Original	Input	0	Pass	Pass	Fail
Original	Default	124	Fail	Fail	Fail
Original	Dataset best	117	Fail	Fail	Fail
Moderate	Input	0	Pass	Pass	Fail
Moderate	Default	121	Fail	Fail	Fail
Moderate	Dataset best	144	Fail	Fail	Fail
Wide	Input	0	Pass	Pass	Fail
Wide	Default	111	Fail	Fail	Fail
Wide	Dataset best	134	Fail	Fail	Fail

The baseline study does not enforce operational_constraints.json during neighbour acceptance. The table audits unchanged loose, medium, and tight scenario requirements after optimisation. An endpoint can pass every maintenance rule and still fail a displacement limit, busy period, or dependency-group requirement. Such an endpoint is a benchmark result, not an approved operational plan.

Displacement is measured against each dataset's own as-loaded checks and includes rebuild effects. Advancing hypothetical maintenance ages to create the moderate and wide inputs is not a legal move from the original dataset. The saved operational requirements are audited without relaxation.

Dataset	Input visible downtime	Best visible downtime	Fixed full downtime
Original	10166	10043	10248
Moderate stagger	11049	11088	11158
Wide stagger	11004	11062	11144

Visible maintenance days can change as legal starts move work across the forecast boundary. Full downtime remains fixed because durations and supplied check counts are preserved. A higher visible mean reflects changed exposure within the fixed window; it does not mean that required maintenance has been removed.

Detailed busy-period and dependency breach counts and displacement checks are retained in each run record. Scenario dates and aircraft groups come from the saved unchanged operational configuration. Operationally constrained tuning should be a separately specified experiment with a chosen scenario enforced during search.

Settings passing the operational audits

Dataset	Scenario	Best audited setting	Worst day	Std deviation	Max displacement
Original	loose	C17	15	1.942977	70
Original	medium	Input	15	1.954052	0
Original	tight	None	NA	NA	NA
Moderate	loose	C17	16	1.705900	69
Moderate	medium	Input	16	1.716837	0
Moderate	tight	None	NA	NA	NA
Wide	loose	C08	20	0.793431	70
Wide	medium	Input	17	1.411357	0
Wide	tight	None	NA	NA	NA

This table filters the measured endpoints by the saved operational requirements and includes the unchanged input when it passes. It answers which tested snapshot is strongest among those that pass every scenario check. It is not a run of an optimiser that enforces those checks during search.

Input means that the unchanged forecast outranks every passing optimisation endpoint in the tested set. None means no tested endpoint or unchanged input passes the scenario. It does not establish mathematical infeasibility of the scenario or show that a constrained search cannot find a better schedule.

Post-search filtering can discard a strong trajectory after it has already taken an unacceptable move. Enforcing displacement limits, busy periods, and dependency groups during proposal generation would produce different accepted paths and potentially different best settings. The primary recommendations in this report apply to the maintenance-only benchmark until that separate constrained study is performed.

Limits of the evidence and reproduction

Dataset	Forced first D overlap	Worst day upper bound	Best measured worst day
Original	6	18	18
Moderate stagger	3	21	19
Wide stagger	3	21	20

Before the first forecast D check, smaller checks do not reset its opening D clock. With a 126-day duration and 70-day early allowance, every legal placement covers the 56-day interval beginning at that D due date. The peak overlap of these forced intervals bounds the worst day at 18 for the original and 21 for each stagger input. A measured result matching its bound proves optimality of the primary objective under those assumptions; otherwise the remaining gap is unresolved.

These bounds do not establish globally optimal smoothness. The study is a broad discrete sweep on three related forecasts, including two synthetic maintenance-age variants, and uses each saved maintenance opening state. It does not establish performance on an independent real fleet, changed maintenance rules, other horizon lengths, or operationally constrained neighbourhoods.

The study retains each supplied check list rather than generating additional check cycles when rebuilds change dates near the forecast boundary. This matches the fixed-list baseline. A planning implementation that regenerates the forecast and adds newly required checks would need a separate verification of the resulting schedules and parameter ranking.

Run .scratch/priority-hyperparameter-analysis/run_three_datasets.py with --dataset original, moderate, or wide and --sweep-only for the independent sweep processes. After every sweep finishes, run finalise_three_datasets.py to obtain serial finalist timings and cap extensions, including the shared setting. The script resumes only against matching input, opening-state, frozen algorithm, frozen adapter, and configuration records.

The per-dataset JSON files store scores, traces, audits, proposal counts, source hashes, and endpoint schedule hashes. Complete schedule JSON files are retained under three-dataset-schedules. The input workbooks, saved opening state, production optimiser, and operational configuration are not edited by the study.

The earlier original-only Word report remains an archive for its longer scoring horizon and small legal start perturbations. Those perturbations are separate from the actual moderate and wide maintenance-age datasets analysed here.

Original full result catalogue

Each setting uses an 800-move sweep cap. A listed 1600 cap indicates a subsequent finalist extension. Local means a full pass found no improving move. The table lists each structural setting once; repeated timings and reference runs remain in the raw records.

ID	Min	Std deviation	Mean	Moves	Cap	End	Days below 19
C01	18	1.422087	21.703492	259	800	Local	142
C02	18	1.404088	21.705090	300	800	Local	116
C03	18	1.410612	21.703721	315	800	Local	129
C04	18	1.408616	21.708514	292	800	Local	133
C05	17	1.654990	21.681123	800	800	Cap	188
C06	17	1.528896	21.700753	800	800	Cap	179
C07	17	1.500076	21.703264	655	800	Local	176
C08	18	1.462199	21.700753	477	800	Local	146
C09	18	1.469950	21.697558	351	800	Local	159
C10	17	1.465157	21.696188	271	800	Local	164
C11	18	1.481357	21.694362	185	800	Local	180
C12	17	1.539122	21.694134	131	800	Local	205
C13	17	1.565118	21.684775	88	800	Local	227
C14	17	1.588813	21.686373	89	800	Local	235
C15	17	1.608533	21.685003	67	800	Local	242
C16	16	1.707803	21.687286	28	800	Local	265
C17	15	1.942977	21.682721	2	800	Local	387
C18	18	1.413456	21.704177	301	800	Local	134
C19	18	1.409795	21.706003	315	800	Local	122
C20	18	1.410969	21.696645	307	800	Local	150
C21	18	1.415264	21.703949	332	800	Local	136
C22	18	1.416652	21.703264	349	800	Local	119
C23	17	1.539088	21.684775	121	800	Local	221
C24	18	1.408017	21.704862	298	800	Local	117
C25	18	1.405517	21.705318	311	800	Local	113
C26	18	1.413299	21.701895	348	800	Local	122
C27	18	1.441412	21.697558	217	800	Local	147
C28	18	1.427091	21.699840	312	800	Local	128
C29	18	1.417940	21.696416	267	800	Local	132
C30	18	1.409722	21.707601	332	800	Local	134
C31	18	1.444352	21.703721	800	800	Cap	118
C32	17	1.654990	21.681123	800	800	Cap	188
C33	18	1.415108	21.700525	345	800	Local	124
C34	17	1.422981	21.705090	772	800	Local	136
C35	18	1.416909	21.703721	237	800	Local	133
C36	18	1.415073	21.703036	248	800	Local	119
C37	18	1.409965	21.703721	294	800	Local	119
C38	18	1.423343	21.699155	289	800	Local	137

ID	Min	Std deviation	Mean	Moves	Cap	End	Days below 19
C39	18	1.417070	21.703721	299	800	Local	132
C40	18	1.427404	21.704405	273	800	Local	133
C41	18	1.428337	21.698927	313	800	Local	149
C42	18	1.407822	21.705090	318	800	Local	118
C43	18	1.406008	21.704177	338	800	Local	118
C44	18	1.409092	21.703036	275	800	Local	123
C45	18	1.411563	21.708286	321	800	Local	144
C46	18	1.421606	21.703492	260	800	Local	136
C47	18	1.417751	21.701210	270	800	Local	152
C48	18	1.413428	21.702123	396	800	Local	124
C49	18	1.411036	21.701895	346	800	Local	139
C50	18	1.413199	21.703721	282	800	Local	124
C51	18	1.408279	21.704177	276	800	Local	121
C52	18	1.407177	21.703949	285	800	Local	121
C53	18	1.413100	21.704405	259	800	Local	123
C54	18	1.450569	21.700982	800	800	Cap	171
C55	17	1.654990	21.681123	800	800	Cap	188
C56	18	1.404088	21.705090	300	800	Local	116
C57	18	1.401929	21.707601	302	800	Local	122
C58	18	1.406948	21.704405	314	800	Local	119

Moderate stagger full result catalogue

Each setting uses an 800-move sweep cap. A listed 1600 cap indicates a subsequent finalist extension. Local means a full pass found no improving move. The table lists each structural setting once; repeated timings and reference runs remain in the raw records.

ID	Min	Std deviation	Mean	Moves	Cap	End	Days below 19
C01	19	0.911708	21.470897	347	800	Local	0
C02	19	0.898502	21.468158	358	800	Local	0
C03	19	0.887635	21.471810	414	800	Local	0
C04	19	0.896507	21.469299	320	800	Local	0
C05	18	1.344982	21.477973	800	800	Cap	4
C06	19	1.170972	21.477973	800	800	Cap	0
C07	19	1.105460	21.471810	702	800	Local	0
C08	18	1.095632	21.468843	493	800	Local	4
C09	19	1.099823	21.469984	370	800	Local	0
C10	17	1.070767	21.468843	272	800	Local	10
C11	18	1.045514	21.468386	243	800	Local	5
C12	18	1.061332	21.468386	177	800	Local	3
C13	18	1.112248	21.471582	129	800	Local	14
C14	17	1.157659	21.469984	103	800	Local	42
C15	18	1.214228	21.469299	74	800	Local	30
C16	17	1.341924	21.477973	41	800	Local	79
C17	16	1.705900	21.477973	2	800	Local	277
C18	19	0.906687	21.470897	367	800	Local	0
C19	19	0.901660	21.471582	315	800	Local	0
C20	19	0.923101	21.469299	347	800	Local	0
C21	19	0.909452	21.470897	380	800	Local	0
C22	19	0.954496	21.470212	405	800	Local	0
C23	18	1.083495	21.469299	158	800	Local	18
C24	19	0.905909	21.470212	329	800	Local	0
C25	19	0.897548	21.469984	351	800	Local	0
C26	19	0.941757	21.470897	352	800	Local	0
C27	19	0.962634	21.471582	274	800	Local	0
C28	19	0.976447	21.469071	360	800	Local	0
C29	19	0.906847	21.468158	345	800	Local	0
C30	19	0.883419	21.469071	367	800	Local	0
C31	19	1.058317	21.468386	800	800	Cap	0
C32	18	1.344982	21.477973	800	800	Cap	4
C33	19	0.936329	21.468386	404	800	Local	0
C34	19	0.946293	21.469071	800	800	Cap	0
C35	19	0.909619	21.468386	328	800	Local	0
C36	19	0.950203	21.470897	296	800	Local	0
C37	19	0.938064	21.469299	293	800	Local	0
C38	19	0.942161	21.468386	354	800	Local	0

ID	Min	Std deviation	Mean	Moves	Cap	End	Days below 19
C39	19	0.963719	21.468386	319	800	Local	0
C40	19	0.923071	21.468386	334	800	Local	0
C41	19	0.898510	21.468386	390	800	Local	0
C42	19	0.903063	21.468158	375	800	Local	0
C43	19	0.897231	21.468158	392	800	Local	0
C44	19	0.909673	21.469984	348	800	Local	0
C45	19	0.901546	21.468158	352	800	Local	0
C46	19	0.923565	21.468386	312	800	Local	0
C47	19	0.905123	21.469299	325	800	Local	0
C48	19	0.910190	21.470441	456	800	Local	0
C49	19	0.885310	21.471582	411	800	Local	0
C50	19	0.916204	21.470897	338	800	Local	0
C51	19	0.903569	21.468158	352	800	Local	0
C52	19	0.899518	21.468158	355	800	Local	0
C53	19	0.905091	21.468386	329	800	Local	0
C54	19	1.064117	21.468158	800	800	Cap	0
C55	18	1.344982	21.477973	800	800	Cap	4
C56	19	0.898502	21.468158	358	800	Local	0
C57	19	0.892207	21.470441	314	800	Local	0
C58	19	0.894978	21.469299	366	800	Local	0

Wide stagger full result catalogue

Each setting uses an 800-move sweep cap. A listed 1600 cap indicates a subsequent finalist extension. Local means a full pass found no improving move. The table lists each structural setting once; repeated timings and reference runs remain in the raw records.

ID	Min	Std deviation	Mean	Moves	Cap	End	Days below 19
C01	20	0.616685	21.473180	313	800	Local	0
C02	19	0.633267	21.476832	309	800	Local	0
C03	20	0.607737	21.473180	362	800	Local	0
C04	20	0.638608	21.475690	297	800	Local	0
C05	18	1.082390	21.488245	800	800	Cap	1
C06	19	0.887747	21.488245	800	800	Cap	0
C07	19	0.841440	21.482082	662	800	Local	0
C08	20	0.793431	21.482995	442	800	Local	0
C09	19	0.805418	21.482766	343	800	Local	0
C10	19	0.792254	21.481853	251	800	Local	0
C11	19	0.778973	21.485962	225	800	Local	0
C12	19	0.797280	21.476375	160	800	Local	0
C13	19	0.843196	21.476375	123	800	Local	0
C14	19	0.913454	21.480027	100	800	Local	0
C15	19	0.946695	21.485962	73	800	Local	0
C16	19	0.978886	21.482766	47	800	Local	0
C17	17	1.363817	21.488245	3	800	Local	81
C18	20	0.606233	21.473180	341	800	Local	0
C19	20	0.603282	21.474777	293	800	Local	0
C20	20	0.603592	21.473180	323	800	Local	0
C21	20	0.613717	21.473180	353	800	Local	0
C22	20	0.655913	21.476375	396	800	Local	0
C23	19	0.809272	21.478430	129	800	Local	0
C24	20	0.611482	21.473180	335	800	Local	0
C25	20	0.609238	21.473180	341	800	Local	0
C26	20	0.633483	21.473180	349	800	Local	0
C27	20	0.667582	21.474777	259	800	Local	0
C28	19	0.684404	21.473180	372	800	Local	0
C29	20	0.637802	21.473408	325	800	Local	0
C30	20	0.586797	21.475006	291	800	Local	0
C31	19	0.749929	21.482082	800	800	Cap	0
C32	18	1.082390	21.488245	800	800	Cap	1
C33	20	0.671271	21.473180	363	800	Local	0
C34	19	0.685575	21.477973	771	800	Local	0
C35	19	0.641371	21.473408	283	800	Local	0
C36	19	0.667404	21.479571	268	800	Local	0
C37	19	0.701177	21.482538	260	800	Local	0
C38	20	0.677062	21.474093	310	800	Local	0

ID	Min	Std deviation	Mean	Moves	Cap	End	Days below 19
C39	19	0.666169	21.473636	324	800	Local	0
C40	19	0.626372	21.476604	304	800	Local	0
C41	20	0.636755	21.474093	314	800	Local	0
C42	20	0.657894	21.473636	276	800	Local	0
C43	20	0.626964	21.473180	321	800	Local	0
C44	20	0.610735	21.473180	325	800	Local	0
C45	20	0.622674	21.475462	288	800	Local	0
C46	20	0.688098	21.474093	237	800	Local	0
C47	20	0.629144	21.473180	282	800	Local	0
C48	20	0.612600	21.473180	414	800	Local	0
C49	20	0.597531	21.473636	378	800	Local	0
C50	20	0.621479	21.473180	311	800	Local	0
C51	19	0.637076	21.473180	296	800	Local	0
C52	19	0.634347	21.476832	308	800	Local	0
C53	19	0.636160	21.477288	298	800	Local	0
C54	19	0.781272	21.473180	800	800	Cap	0
C55	18	1.082390	21.488245	800	800	Cap	1
C56	19	0.633267	21.476832	309	800	Local	0
C57	20	0.608959	21.475462	316	800	Local	0
C58	20	0.650567	21.473636	298	800	Local	0

Complete parameter catalogue

Neighbourhoods are ordered from left to right and separated by a vertical bar. Offsets within a neighbourhood are in their tested order and measured in days. Selection is part of the structural setting, so identical offsets with different policies retain separate IDs.

ID	Selection	Ordered offsets in days
C01	first level	-28,+28 | -14,+14 | -7,+7 | -3,+3
C02	first level	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
C03	first level	-28,+28 | -21,+21 | -14,+14 | -7,+7 | -3,+3 | -1,+1
C04	first level	-42,+42 | -28,+28 | -14,+14 | -7,+7 | -3,+3
C05	first level	-1,+1
C06	first level	-2,+2
C07	first level	-3,+3
C08	first level	-5,+5
C09	first level	-7,+7
C10	first level	-10,+10
C11	first level	-14,+14
C12	first level	-21,+21
C13	first level	-28,+28
C14	first level	-35,+35
C15	first level	-42,+42
C16	first level	-56,+56
C17	first level	-70,+70
C18	first level	-28,+28 | -14,+14 | -7,+7 | -3,+3 | -1,+1
C19	first level	-56,+56 | -28,+28 | -14,+14 | -7,+7 | -3,+3
C20	first level	-70,+70 | -42,+42 | -28,+28 | -14,+14 | -3,+3
C21	first level	-28,+28 | -14,+14 | -7,+7 | -1,+1
C22	first level	-14,+14 | -7,+7 | -3,+3 | -1,+1
C23	first level	-70,+70 | -42,+42 | -28,+28
C24	first level	-32,+32 | -16,+16 | -8,+8 | -4,+4 | -2,+2
C25	first level	-32,+32 | -16,+16 | -8,+8 | -4,+4 | -2,+2 | -1,+1
C26	first level	-28,+28 | -7,+7 | -1,+1
C27	first level	-28,+28 | -7,+7
C28	first level	-14,+14 | -3,+3
C29	first level	-30,+30 | -15,+15 | -7,+7 | -3,+3
C30	first level	-56,+56 | -42,+42 | -28,+28 | -14,+14 | -7,+7 | -3,+3 | -1,+1
C31	first level	-3,+3 | -7,+7 | -14,+14 | -28,+28
C32	first level	-1,+1 | -7,+7 | -14,+14 | -28,+28
C33	first level	-14,+14 | -28,+28 | -3,+3 | -7,+7
C34	first level	-42,+42 | -1,+1 | -28,+28 | -7,+7
C35	first level	-28,-14,+14,+28 | -7,-3,+3,+7
C36	first level	-28,-14,-7,-3,+3,+7,+14,+28
C37	first level	-28,-21,-14,-7,-3,-1,+1,+3,+7,+14,+21,+28
C38	first level	-28,-14,+14 | -7,-3,+3

ID	Selection	Ordered offsets in days
C39	first level	-14,+14,+28 | -3,+3,+7
C40	all levels	-28,+28 | -14,+14 | -7,+7 | -3,+3
C41	first improving	-28,+28 | -14,+14 | -7,+7 | -3,+3
C42	all levels	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
C43	all levels	-28,+28 | -21,+21 | -14,+14 | -7,+7 | -3,+3 | -1,+1
C44	all levels	-32,+32 | -16,+16 | -8,+8 | -4,+4 | -2,+2 | -1,+1
C45	all levels	-56,+56 | -42,+42 | -28,+28 | -14,+14 | -7,+7 | -3,+3 | -1,+1
C46	all levels	-28,-14,+14,+28 | -7,-3,+3,+7
C47	all levels	-42,+42 | -28,+28 | -14,+14 | -7,+7 | -3,+3
C48	first improving	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
C49	first improving	-28,+28 | -21,+21 | -14,+14 | -7,+7 | -3,+3 | -1,+1
C50	first level	-28,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
C51	first level	-28,-14,+14,+28 | -14,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
C52	first level	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,+7 | -3,-1,+1,+3
C53	first level	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,+3
C54	first level	-3,-1,+1,+3 | -7,-3,+3,+7 | -14,-7,+7,+14 | -28,-14,+14,+28
C55	first level	-1,+1 | -3,+3 | -7,+7 | -14,+14 | -21,+21 | -28,+28
C56	first level	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3 | -2,+2
C57	first level	-42,+42 | -28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
C58	first level	-28,-21,-14,+14,+21,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3

Source record

Input or code	SHA256
original	a98a1a3725cf3da075034932aa94e314 d8b99933e2b942b305730a35978667a2
moderate	7bd84e57df88ede02715cbd61ac2e6da a756abdc00f9624981f1b20b4a122e99
wide	41d5e09fe44c13833324d58b3f9dcb69 f4bac71011733373efdb1aa373cb3096
opening_state	63bf4f7f36465701eebf9c7e725d8cc2 fdef334b49c29671dcbdc4b5e63425d3
Frozen algorithm	0ff76a38ca83a163376564ffaee5cfd7 192ecbdcfd8d5371ecd7cd1b2ff508ea
Frozen adapter	f6c1b0caf360a6bc4b7f125ac7ff5b81 d63ebf801fa3e556bd05824099c3f5d7
Final study driver	df46fd54afe6fad5c46d7bf8550383dc afd0cfab0255b67f104ee65001802bdb
Operational rules	91aa0eecb412d7f4ad3a00a1132a5978 bad0958b86723839defb75cea8a4bbc9

Original input heavy_maintenance_forecast.xlsx

Moderate input outputs/initial-stagger-20261013/heavy_maintenance_forecast_stagger_moderate.xlsx

Wide input outputs/initial-stagger-20261013/heavy_maintenance_forecast_stagger_wide.xlsx

Opening state .scratch/initial-maintenance-stagger/schedule-data.json

Frozen opening adapter .scratch/priority-hyperparameter-analysis/frozen-source/run_priority_optimise.py

Frozen algorithm .scratch/priority-hyperparameter-analysis/frozen-source/savn_priority_based_selection.py

The frozen algorithm is the byte-identical CRLF form of Git revision f9459f4. The current live entry point is vn_priority_based_selection.py; the hashes above identify the version actually evaluated.

Study driver .scratch/priority-hyperparameter-analysis/run_three_datasets.py

three-dataset-original.json
SHA256 c5c09d20306004a56023dd4bda9aecde
9184f6ff2e41b6796bdddba0d5eeabef

three-dataset-moderate.json
SHA256 7959d5556af7cbe99fc4246c20cb3779
ba26e4918acb7747ae9a6cf38a0ed368

three-dataset-wide.json
SHA256 e040346b33e5a93580f658b7e44d1575
2bee4b6fdd05bb29c327f41bfeae61b4